

Paper 2 - Set 10 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	D	Functions and transformations
2	B	Sequences and series
3	A	Polynomials, factor and remainder theorems
4	E	The logic of arguments
5	B	Chains of reasoning
6	A	Circles and circle theorems
7	A	Identifying errors in proofs
8	A	The logic of arguments
9	D	The logic of arguments
10	D	Chains of reasoning
11	E	Trigonometric functions and equations
12	F	Sequences and series
13	A	Mathematical proof
14	E	GCSE algebra
15	F	Functions and transformations
16	A	Chains of reasoning
17	A	Identifying errors in proofs
18	E	Mathematical proof
19	C	Differentiation
20	D	Indices and surds

Worked solutions

Question 1 FUN-14 · Functions and transformations · spec MM8.2

The graph of $y = f(x)$ has a maximum point at $(3, 8)$. The graph of $y = f(x - a) + b$ has its maximum at $(1, 2)$. Find $a + b$.

Answer: D

The maximum moves from $(3, 8)$ to $(3 + a, 8 + b) = (1, 2)$, so $a = -2$, $b = -6$ and $a + b = -8$.

Question 2 SEQ-09 · Sequences and series · spec MM2.2

How many terms of the arithmetic series $3 + 7 + 11 + \dots$ are needed for the sum to first exceed 500?

Answer: B

$S_n = \frac{n}{2}(6 + 4(n - 1)) = n(2n + 1)$. $n = 15$: 465; $n = 16$: 528 > 500. So 16.

Question 3 POL-11 · Polynomials, factor and remainder theorems · spec MM1.6

Find the coefficient of x^2 in the expansion of $(x + 1)(x - 2)(2x + 3)$.

Answer: A

$(x + 1)(x - 2) = x^2 - x - 2$; multiply by $(2x + 3)$: the x^2 term is $3x^2 - 2x^2 = x^2$, coefficient 1.

Question 4 LGC-13 · The logic of arguments · spec Arg4

Which of the following is the negation of $x > 3$ and $y \leq 5$?

Answer: E

Not $(A \text{ and } B)$ is $(\text{not } A) \text{ or } (\text{not } B)$. Not $x > 3$ is $x \leq 3$; not $y \leq 5$ is $y > 5$.

Question 5 RSN-07 · Chains of reasoning · spec Prf5

The digits 1 to 9 are placed in a 3×3 grid so that every row and every column has the same sum. What is that sum?

Answer: B

The three rows together contain every digit once, so $3 \times (\text{row sum}) = 1 + 2 + \dots + 9 = 45$, giving 15.

Question 6 X2-10 · Circles and circle theorems · spec MM3.2

The circle C has centre (a, b) and radius r . P : ' $r > |b|$ '. Q : ' C crosses the x -axis at two distinct points'. Which one of the following is true?

Answer: A

The distance from the centre to the x -axis is $|b|$; the circle cuts the axis twice exactly when this is less than r .

Question 7 ERR-04 · Identifying errors in proofs · spec Err1

Claim: for all real x , if $x^2 > 9$ then $x > 3$.

- I Suppose $x^2 > 9$.
- II Taking square roots of both sides, $x > 3$.
- III This proves the claim.

Which of the following is true?

Answer: A

$x^2 > 9$ gives $|x| > 3$, i.e. $x > 3$ or $x < -3$. The claim itself is false, and line II is where the invalid deduction happens.

Question 8 LGC-01 · The logic of arguments · spec Arg1

Consider the statement: ****if n is a multiple of 6, then n is a multiple of 3.**** Which of the following is its contrapositive?

Answer: A

The contrapositive of 'if A then B ' is 'if not B then not A '. It is logically equivalent to the original; B is the converse and C is the inverse.

Question 9 LGC-14 · The logic of arguments · spec Arg1

For a positive integer n , consider the statement: ****if n is prime, then n is odd.**** Which one of the following is true?

Answer: D

$n = 2$ is an even prime, so the statement is false. The converse, 'if n is odd then n is prime', fails for $n = 9$.

Question 10 RSN-22 · Chains of reasoning · spec Prf5

A 4×4 grid of squares is coloured so that every row and every column contains exactly two black squares. How many black squares are there in the grid?

Answer: D

Each of the 4 rows contains exactly 2 black squares: 8 in total.

Question 11 X2-07 · Trigonometric functions and equations · spec MM4.6

Consider the equation $\sin x = k$ where k is a real constant. Which of the following statements is/are true?

- I For every k with $-1 < k < 1$ the equation has exactly two solutions in $0 \leq x < 2\pi$.
- II For $k = 1$ the equation has exactly one solution in $0 \leq x < 2\pi$.
- III For every real k the equation has at least one solution.

Answer: E

I and II are read from the graph of $\sin$ over one period. III is false: $\sin x = 2$ has no solutions.

Question 12 SEQ-12 · Sequences and series · spec MM2.3

Which of the following geometric series is/are convergent?

I $1 + 0.9 + 0.81 + \dots$

II $3 - 3 + 3 - 3 + \dots$

III $100 + 10 + 1 + 0.1 + \dots$

Answer: F

A geometric series converges if and only if $|r| < 1$. I has $r = 0.9$, III has $r = 0.1$: both converge. II has $r = -1$ and does not converge.

Question 13 PRF-12 · Mathematical proof · spec Prf2

For a positive integer n it is known that: (i) if n is a multiple of 4 then n is not a multiple of 6; (ii) n is a multiple of 12 or n is a multiple of 4. What can be deduced?

Answer: A

If n were a multiple of 12 it would be a multiple of both 4 and 6, contradicting (i). So by (ii) n is a multiple of 4, and by (i) not a multiple of 6.

Question 14 ALG-08 · GCSE algebra · spec M4.19

The n th term of a sequence is $\frac{n^2 + 1}{2n}$. Which of the following statements is/are true?

I The 5th term is 2.6.

II Every term after the first is greater than the previous term.

III Some term of the sequence equals 10.

Answer: E

I: $\frac{26}{10} = 2.6$. II: $u_n = \frac{n}{2} + \frac{1}{2n}$; $u_1 = 1$, $u_2 = 1.25$, and for $n \geq 1$ the increase $\frac{1}{2} - \frac{1}{2n(n+1)} > 0$, so increasing. III: $n^2 - 20n + 1 = 0$ has no integer solutions (discriminant 396 is not a square), so false.

Question 15 X2-11 · Functions and transformations · spec MM8.2

Let $f(x) = x^2$ for all real x , and let g be the function whose graph is obtained by translating the graph of $y = f(x)$ by 2 units in the positive x -direction and then reflecting in the y -axis. Which of the following statements is/are true?

I $g(x) = (x + 2)^2$

II $g(x) = (x - 2)^2$

III The graph of g has its vertex at $(-2, 0)$

Answer: F

Translation gives $(x - 2)^2$; reflecting in the y -axis replaces x by $-x$: $(-x - 2)^2 = (x + 2)^2$, with vertex at $(-2, 0)$.

Question 16 RSN-17 · Chains of reasoning · spec Prf5

Alice, Ben and Cara each make one statement. Alice: 'Ben is lying.' Ben: 'Cara is lying.' Cara: 'Alice and Ben are both lying.' Exactly one of them is telling the truth. Who is it?

Answer: A

If Alice is truthful then Ben lies, so Cara is truthful: two truth-tellers, impossible. If Cara is truthful then Alice and Ben both lie; but Alice lying means Ben is truthful: contradiction. So Ben is the truth-teller: Cara lies (indeed Ben is not lying) and Alice lies (Ben is not lying), which is consistent.

Question 17 ERR-07 · Identifying errors in proofs · spec Err2

A student writes the following argument.

I $0.1 > 0.01$.

II Taking logarithms to base 10 of both sides: $-1 > -2$.

III Multiplying both sides by $\log_{10} \frac{1}{2}$, which is negative, reverses the inequality:
 $-\log_{10} \frac{1}{2} < -2\log_{10} \frac{1}{2}$.

Which of the following is correct?

Answer: A

Logarithms are increasing functions, so line II is valid; $\log_{10} \frac{1}{2} < 0$, so reversing the inequality in line III is correct.

Question 18 PRF-18 · Mathematical proof · spec Prf2

Every element of a set S of integers is either a multiple of 3 or a multiple of 5. Every multiple of 3 in S is negative. S contains the number 10. Which of the following statements must be true?

I S contains a positive multiple of 5.

II S contains no positive multiple of 3.

III S contains a negative number.

Answer: E

$10 \in S$ is a positive multiple of 5 (I). Multiples of 3 in S are negative, so none is positive (II). III fails: $S = \{10\}$ satisfies everything.

Question 19 X2-02 · Differentiation · spec MM6.3

Let f be a polynomial. P : ' $f'(x) > 0$ for all real x '. Q : ' $f(x) = 0$ has exactly one real root'. Which one of the following is true?

Answer: C

If $f' > 0$ everywhere, f is strictly increasing and (being an odd-degree polynomial) crosses the axis exactly once: P is sufficient. But $f(x) = x^3$ has exactly one root and $f'(0) = 0$, so P is not necessary.

Question 20 IND-12 · Indices and surds · spec MM1.1

For how many integer values of n with $1 \leq n \leq 100$ is $\sqrt{n} + \sqrt{n+1}$ rational?

Answer: D

For the sum to be rational, both $\sqrt{n}$ and $\sqrt{n+1}$ would need to be rational (if one is irrational, $\sqrt{n} + \sqrt{n+1}$ squared gives $2n+1 + 2\sqrt{n(n+1)}$ and $n(n+1)$ is never a perfect square for $n \geq 1$ since $n^2 < n(n+1) < (n+1)^2$).

No two consecutive positive integers are both squares, so the answer is 0.